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VARIABLE:

Variables is a primary process of holding a values or datas in the single name. Variables are process or identific has a keyword.

Keywords are var, let and const.

Ex: let name = "Anu"

let age = 20;

→ name is a variable that stores "Anu".

→ Age is a variable that stores 20.

Different Between.

let	var	const
let is used to declare a variable when the value may change. Ex: let age = 20; age = 22; console.log(age);	var is used to declare a variable in javascript Ex: var name = "Anu"; name = "Priya"; console.log(name);	const is used to declare a variable when the value should not be reassigned. Ex: const country = "India"; console.log(country);

2. JavaScript Data Types:

Data type is actual value for a programming all values or datas have a type. based on type the process will change. Every data type has a predefined type.

There are two types:

→ Primitive data type

→ Non-Primitive data type.

String:

A string is used to store text or characters. strings are written inside single quotes (' '), double quotes (" ").

```
Ex: let name = "Haru";  
    console.log(name);
```

number:

A number is used to store numeric values, including whole number and decimal numbers.

```
Ex: let age = 22;  
    console.log(age);
```

Boolean:

A Boolean represents one of two values, true or false.

```
Ex: let student = true;  
    console.log(student);
```

Undefined:

Undefined means a variable has been declared but no value has been assigned to it.

```
Ex: let city;  
    console.log(city);
```

Null:

Null represents an intentionally empty or unknown value.

```
Ex: let phoneNumber = null;  
    console.log(phoneNumber);
```

3. Different Between:

==	===
The == operator compares the values of two variables.	The === operator compares both the value and data type.
It can convert the data type before comparing	It does not convert the data type.
Ex: let a=10; let b="10"; console.log(a==b); output: true	Ex: let a=10; let b="10"; console.log(a===b); output: false

4. Operators with Examples:

1. Arithmetic Operators:

Arithmetic operators are used to perform mathematical operations.

operator	meaning	Example
+	Addition	$10 + 5 = 15$
-	Subtraction	$10 - 5 = 5$
*	Multiplication	$10 * 5 = 50$
/	Division	$10 / 5 = 2$
%	Modulus	$10 \% 3 = 1$

2. Assignment Operators

Assignment operators are used to assign or update values in variables.

operator	Meaning	Example
=	Assign	$a = 10$
+=	Add and assign	$a += 5$
-=	Subtract and "	$a -= 5$

$*$ =	Multiply and assign	$a* = 5$
$/$ =	Divide and assign	$a/ = 5$

3. Comparison Operators:

Comparison operators are used to compare two values. They return either true or false.

operator	Meaning	Example
$=$	Equal to	$10 = "10" \rightarrow \text{true}$
$===$	Strictly Equal	$10 === "10" \rightarrow \text{false}$
$!=$	Not Equal	$10 != 5 \rightarrow \text{true}$
$!==$	Strictly not Equal	$10 !== "10" \rightarrow \text{true}$
$>$	Greater than	$10 > 5 \rightarrow \text{true}$
$<$	less than	$10 < 5 \rightarrow \text{false}$
$>=$	Greater than or Equal	$10 >= 10 \rightarrow \text{true}$
$<=$	less than or Equal	$5 <= 10 \rightarrow \text{true}$

logical operators:

Logical operators are used to combine or check multiple conditions.

operator	Meaning	Example
$\&\&$	AND	$\text{true} \&\& \text{true} \rightarrow \text{true}$
$!$	NOT	$!\text{true} \rightarrow \text{false}$

5. Difference Between.

Array	Object	Array of Objects
An array is used to store multiple in a single variable.	An object is used to store related information using key-value pairs.	An array of objects is an array that contains multiple objects.
Array values are ordered and each value has an index value	for example instead of creating separate variable for a student's name age course	Each object can store different information using key-value pairs.

Syntax:

```
let array name = [Value1, Value2, Value 3];
```

Ex:

```
let fruits = ["Apple", "Mango"];  
console.log(fruits);
```

Index:

Apple → 0

Mango → 1

Syntax:

```
let object name = {  
  key: Value,  
  key: Value  
};
```

Ex: let Student = { name: "Arun",
 age: 22,
 course: "JS"
};

console.log(Student);

Syntax:

```
let array name = {  
  {  
    key1: Value 1,  
    key2: Value 2.  
  }  
}
```

Ex: let Product = {
 {

name: "Shirt",

Price: 1000

};

console.log(Product);

6. function:

A function is a reusable block of code used to perform a specific task. Instead of writing the same code multiple times, we can create a function once and call it whenever needed.

Ex: function greeting() {
 console.log("Hello JavaScript");
}
greeting();

output:

Hello JavaScript

⇒ Arguments:

Arguments are the actual values passed when calling a function.

Ex:

```
function greeting(name) {  
  console.log("Hello" + name);  
}  
greeting("Arun");
```

⇒ Parameters:

Parameters are variable written inside the function definition. They are used to receive values.

Ex:

```
function greeting (name) {  
  console.log ("Hello " + name);  
}  
greeting ("Arun");
```

Here:

name $\rightarrow$ is a parameter.

$\Rightarrow$ Return:

return is used to send a value back from a function.

Ex: function add (a, b) {

return a+b;
}

let result = add (10, 20);

console.log(result);

output:

30.

PART-B

BASIC PROGRAMS:

7. Even Or Odd

let number = 27;

if (number % 2 === 0) {

console.log ("Even");

} else {

console.log ("odd");

}

output:

odd

8. For loop:

Print all Even number from 1 to 50.

```
for (let i=1; i<=50; i++){  
  if (i%2==0){  
    console.log(i);  
  }  
}
```

Output:

2
4
6
8
⋮
50

9. factorial:

```
let number=5;  
let factorial=1;  
for (let a=1; a<=number; a++){  
  factorial=factorial*a;  
}  
console.log(number+"!=" , factorial);
```

Output:

5! = 120

10. FIBONACCI SERIES:

```
let a=0;  
let b=1;  
for (let i=1; i<=8; i++){  
  console.log(a);  
  let next = a+b;  
  a=b;  
  b=next;  
}
```

Output:

0
1
1
2
3
5
8
13

PART - c Array & Object:

11.

```
let number = [10, 15, 20, 25, 30, 35, 40];  
for (let a=0; a < number.length; a++) {  
  if (number[a] % 2 == 0) {  
    console.log(number[a]);  
  }  
}
```

output:

10
20
30
40

12.

```
let Marks = [80, 75, 90, 85, 70];  
let total = 0;  
for (let a=0; a < mark.length; a++) {  
  total = total + marks[a];  
}  
console.log("Total=", total);
```

output

Total = 400

13. Object:

```
let Student = {
```

```
  name: "Arjun",
```

```
  age: 23,
```

```
  course: "full stack",
```

```
  Mark: 85
```

```
};
```

```
console.log("Student Name:" + Student.name);
```

```
console.log("Course: " + Student.course);
```

```
console.log("Mark: " + Student.Mark);
```

Output:

Student name: Arjun

Course: full stack

Mark: 85

14. Array Of Objects:

```
let students = [
```

```
  {name: "Arun", Mark: 80},
```

```
  {name: "Kumar", Mark: 45},
```

```
  {name: "Priya", Mark: 90},
```

```
  {name: "Ram", Mark: 35}
```

```
];
```

```
for (let a = 0; a < students.length; a++) {
```

```
  if (students[a].Mark >= 50) {
```

```
    console.log(students[a].name, students[a].Mark);
```

```
  }
```

```
}
```

Output:

Arun 80

Priya 90

PART- D function & Logic

15. function - Largest Number

```
function findLargest(a,b){
```

```
  if(a > b){
```

```
    return a;
```

```
  } else {
```

```
    return b;
```

```
  }
```

```
}
```

```
console.log(findLargest(25,40));
```

Output:

40